Cisco

Networking Project 2025

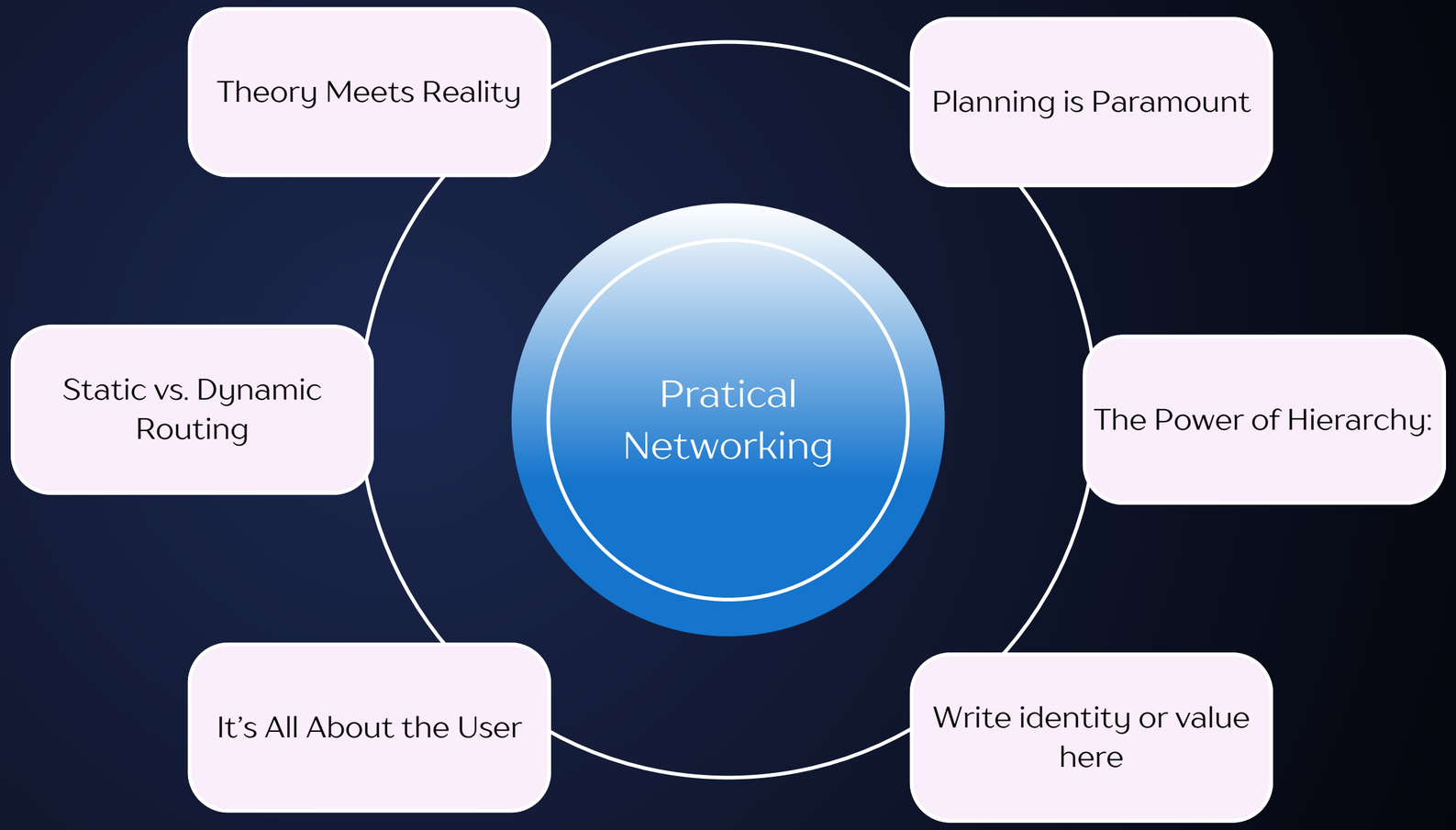
Building & Learning in Distributed Systems

What We Built

A model of a realistic, scalable network for a small enterprise

Key Takeaways & Learnings

This project was a masterclass in practical networking



Architecture

Central Headquarters

with its own data centerhosting critical services (DNS, DHCP, HTTP, FTP)

Two remote Branch Offices

each requiring secure and reliable access to HQ resources

ISP router

providing a gateway to the Internet

Hierarchical switch design

Core, Distribution, Access layers for optimal traffic flow

Architecture

Structured IP addressing plan

using VLSM to efficiently allocate address space across different network segments

VLANs

to logically segment users, servers, enhancing both security and performance

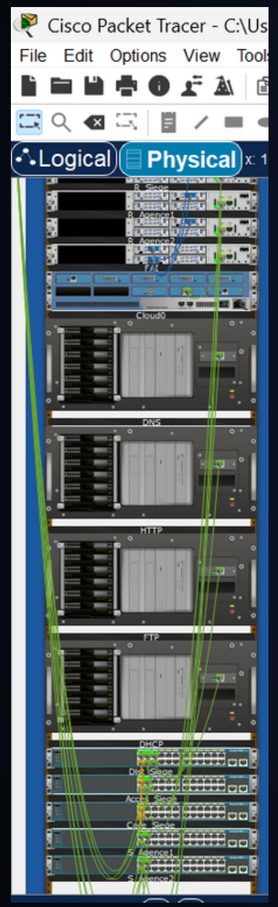
ISP router

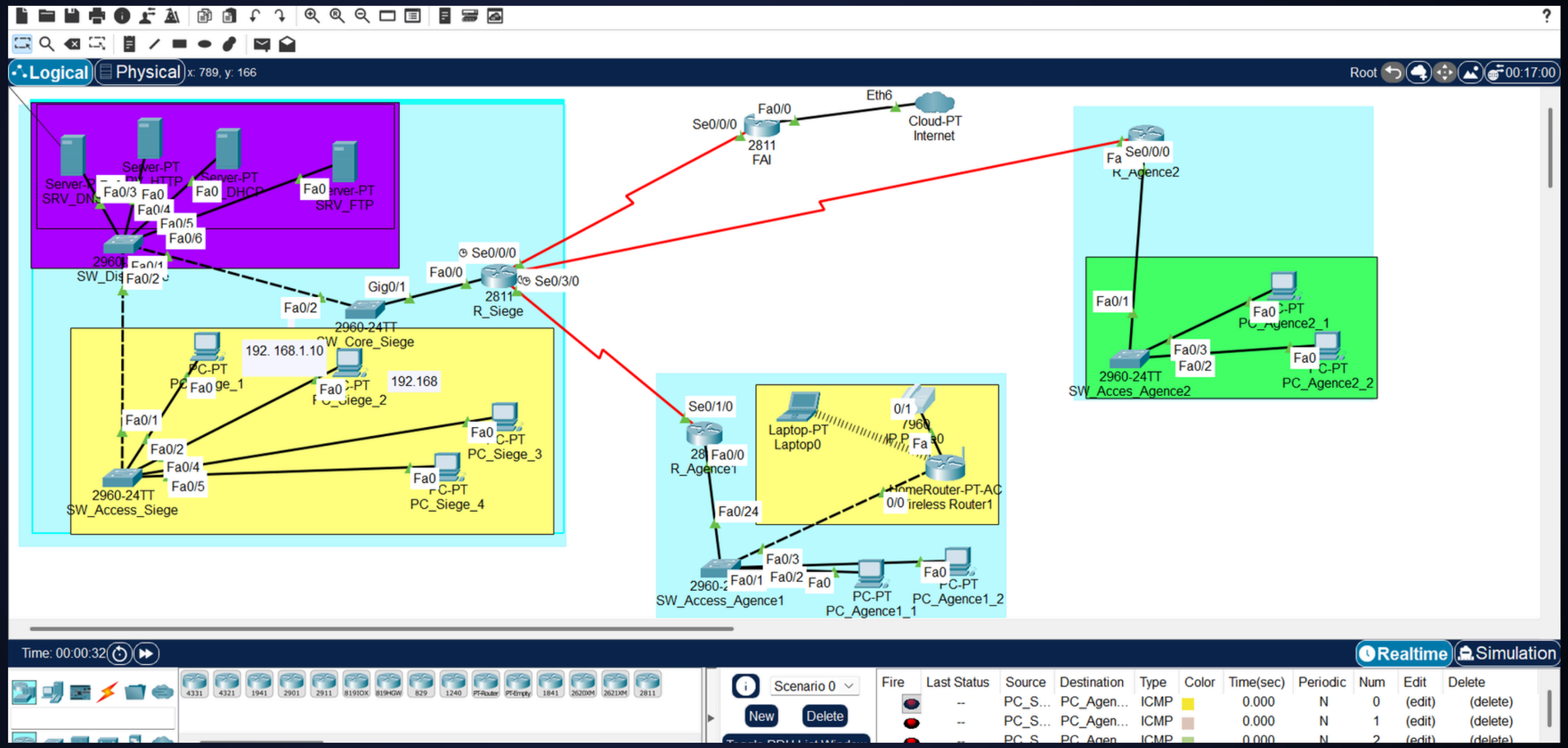
providing a gateway to the Internet

Access Control Lists (ACLs)

Basic security policies implemented to control inter-VLAN and Internet-bound traffic

Visualisation





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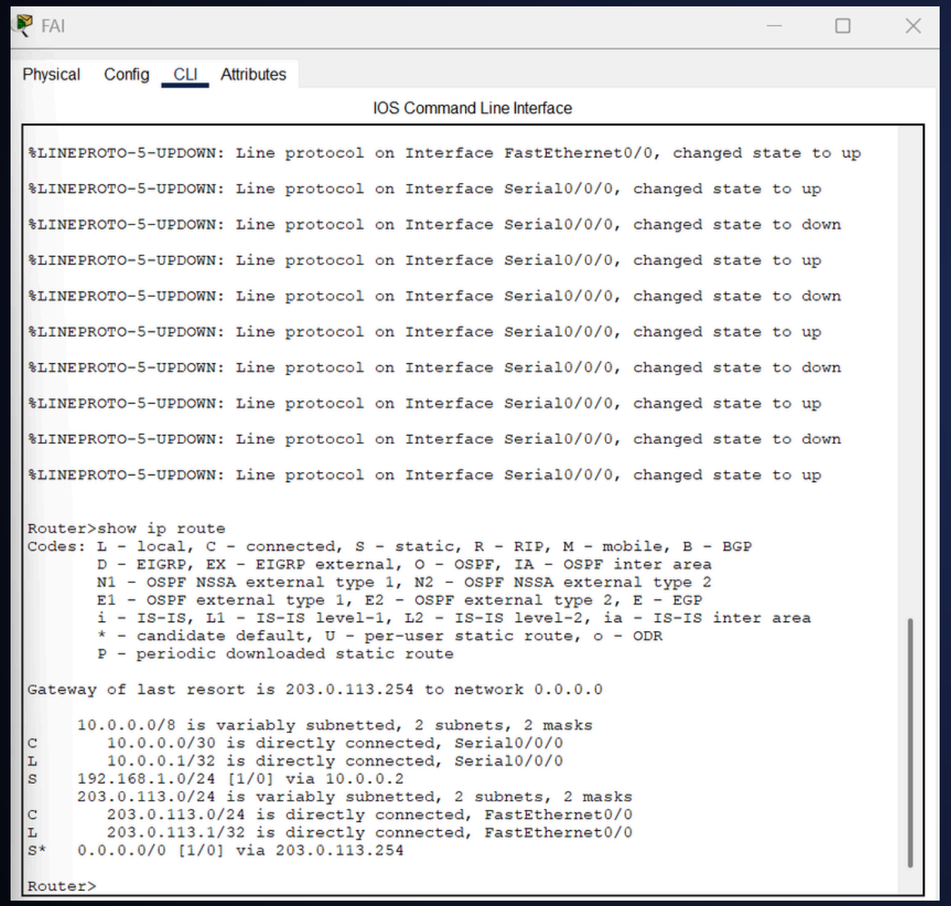

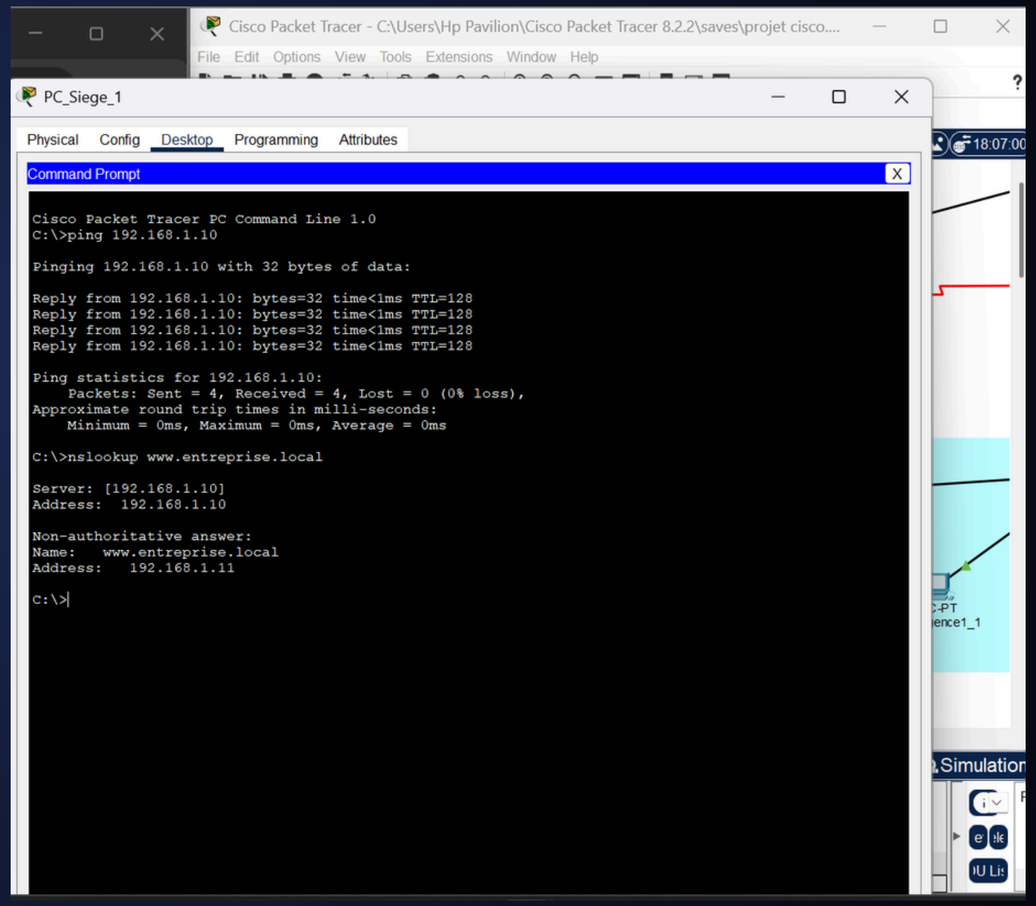

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